DATASCI 406 FA 26, Quiz 3.4

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

Each Flint home gives a difference D_i, first draw minus flushed. The sign-flip test uses H_0: each D_i is symmetric about 0. What does the test condition on, and what does it rearrange?

1. Conditions on $|D_i|$; flips the signs, all patterns equally likely.
2. Conditions on $|D_i|$; makes each sign + with the observed fraction of positive D_i.
3. Conditions on the first-draw values; shuffles the flushed values among homes.
4. Conditions on nothing; resamples the D_i with replacement.

Question 2 (1 point)

A two-sample permutation test of $H_0 : F = G$ uses the difference in means. A student keeps the same label shuffles and computes the difference in medians instead. Which is true of the new test?

1. Under $F = G$ it still rejects at most 5% of the time; its power can change.
2. Under $F = G$ it no longer rejects 5% of the time, since the median has no simple null distribution.
3. It gives the same p-value, since the shuffles are the same.
4. It tests equal medians, so it rejects 5% of the time whenever the medians are equal.

Question 3 (1 point)

Before computers, books printed tables of the rank-sum test's null distribution, but not of the permutation null of the difference in means. Why could the rank-sum null be tabulated?

1. Ranks are approximately normal, so one normal table serves every sample.
2. Without ties, its null depends only on m and n, not on the data.
3. The rank sum is always symmetric about 0, so half the table suffices.
4. The difference in means needs σ^2, which a table cannot know.

Question 4 (1 point)

In Part D.3, casual riders on the 231 other days had an SD of about 870, against 390 on the 500 working days. After centering each group at 0, the permutation test of the difference in means rejected 14% of the time at 5%, and Welch's statistic about 6%. Why does the difference in means reject too often?

1. $B = 400$ shuffles is too few, so the p-values are too small.
2. Casual riders are skewed, so means are less stable than medians.
3. Shuffling pools the two spreads, so the permutation SE is too small when the smaller group has the larger spread.
4. Centering each group made the means unequal, so H_0 was false.